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A MINI PROJECT ON

FLOOD PREDICTION IN KERALA

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UNDER THE GUIDANCE OF

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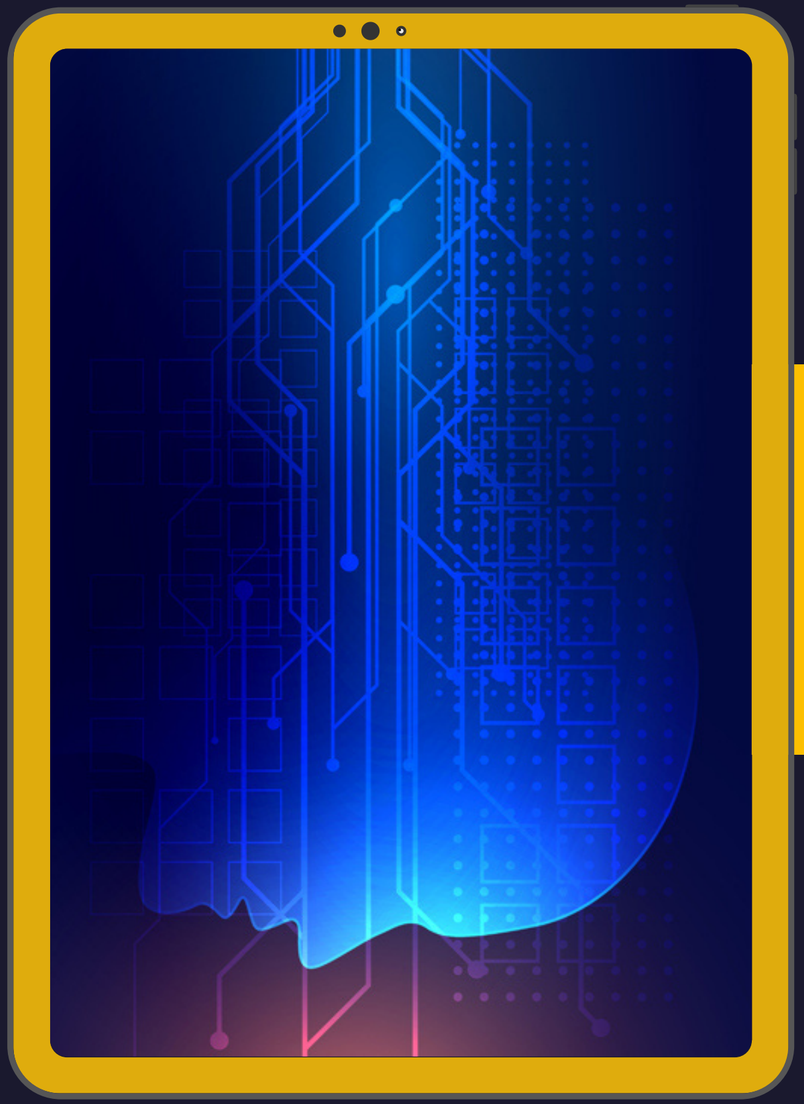


Abstract

A flood is an overflow of water that submerges land that is usually dry. Flood may occur when there's an overflow of water bodies. Kerala has been experiencing floods due to various weather conditions and patterns. Unexpected above normal downpour, extreme rainfall events occurring almost across Kerala during the season, over 90% reservoir storage even before the onset of extreme rainfall events, and finally, the unprecedented extreme rainfall in the catchment areas of major reservoirs in the state lead to the disaster. The main motive of this project is to predict the flood occurrence in the future based on the previous year’s flood rate.

We will be using five algorithms to predict the outcomes:

1. Support vector classifier
2. K-nearest neighbors
3. Random forest
4. Decision tree.
5. Logistic regression

Graphical user interface (GUI) also used for the I/p’s to be considered.

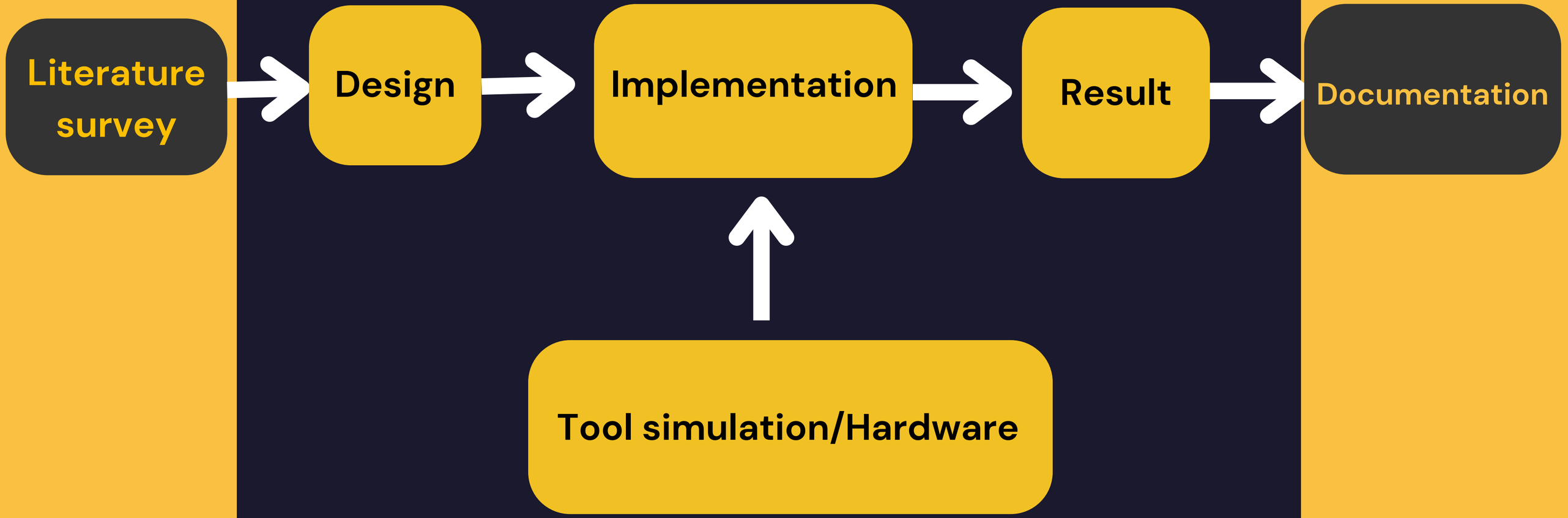
INTRODUCTION

- Kerala has been experiencing sever floods over past few decades. It is possible to predict floods in Kerala through machine learning. We can use previous data sets and weather patterns to predict floods in kerala.

- Two main methods i.e the logistic regression and super vector machine are used to find the time of flood and are compared for accurate results.

- Graphical user interface is used for exhibiting the output. The result is a categorical result expressing if the flood is likely to occur or not.

PROJECT WORKFLOW



TOOLS REQUIRED

Pandas

NumPy

Tkinter

Python Pickle

Seaborn

Matplotlib

PLAN OF ACTION

Week – 1

Analyzing probelm, gathering previous data sets, making a roadmap for perfect model.

Week – 2

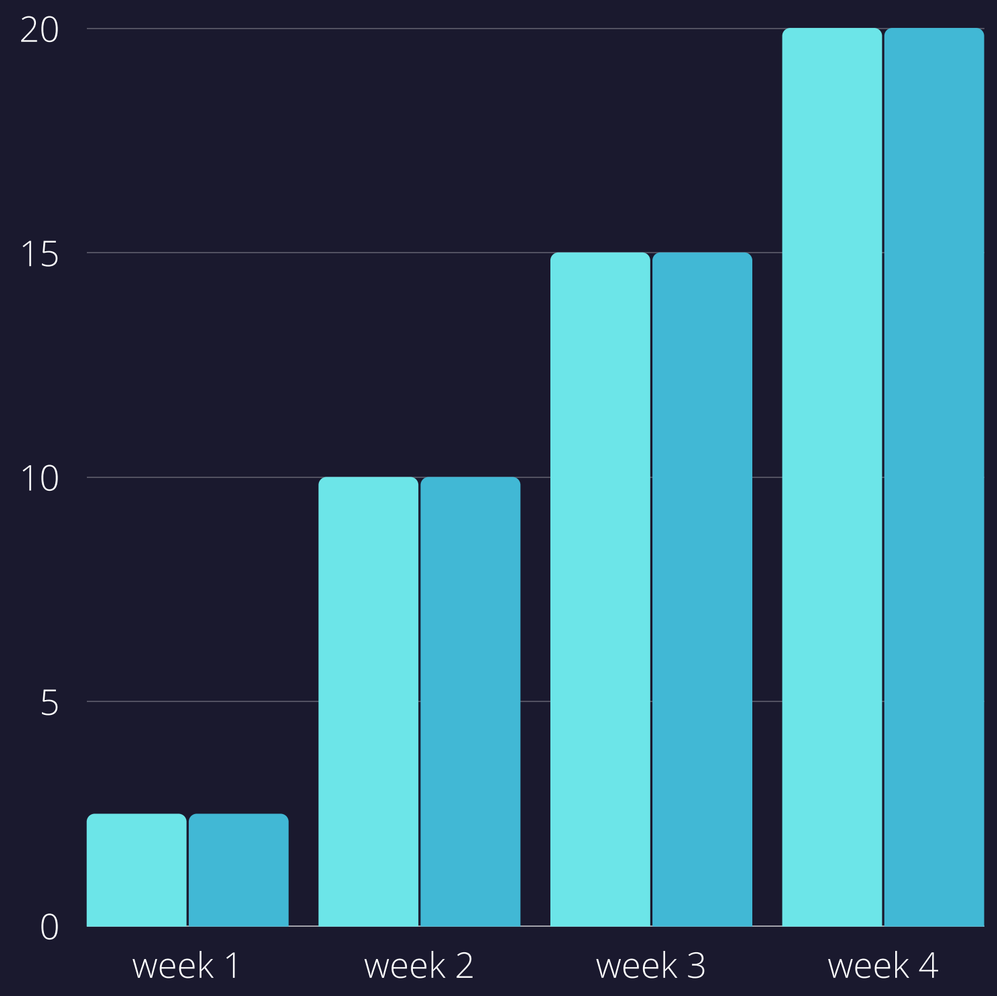
Comparing algorithms, writing the source code.

Week – 3

Employing libraries and completing the source code.

Week – 4

Developing a aesthetic graphical user interface for the outputs.



PREVIOUS METHOD

Logistic Regression

Linear Regression is one of the most used Machine Learning algorithm.

In regression analysis, logistic regression is estimating the parameters of a logistic model

Logistic regression may be used to predict the risk of floods based on observed characteristics of the weather patterns.

PROPOSED SYSTEM

- To obtain better results we will use logit regression and svm.
- Logit regression can interpret model coefficients as indicators of feature importance.
- SVR gives us the flexibility to define how much error is acceptable in our model and will find an appropriate line.
- The results are compared and accurate result is exhibited.

References

https://www.ijert.org/research/flood-prediction-using-logistic-regression-for-kerala-state-IJERTCONV9IS03010.pdf

https://iopscience.iop.org/article/10.1088/1742-6596/1367/1/012087/pdf

https://github.com/rajatkeshri/Flood-prediction

SUGGESTIONS